AI KAISE SOCHTA HAI

Anthropic ki research, aasaan Hinglish mein. Reel ka poora breakdown: kya mila, kaise mila, aur kitna bharosa karein.



PEHLE YE JAAN LO

- **Research kisne ki:** Anthropic ki interpretability team. Unhone model ke andar ke "features" trace kiye, yaani dekha ki jawab dene se pehle andar kya activate hota hai.
- **Kaun sa model:** deep case studies **Claude 3.5 Haiku** par hui hain. Reel mein "Claude" isi research ke liye bola hai, har AI ke liye nahi.
- **Reel ka ek hi sawaal:** kya AI sirf agla word guess karta hai? Jawab: sirf itna nahi. Neeche 6 saboot.

EK LINE MEIN

Model likhne se pehle aage ka plan bana sakta hai, beech ke steps jod sakta hai, aur ek hi idea ko kai languages mein use karta hai. Par jo tareeka wo khud batata hai, wo hamesha uska asli tareeka nahi hota.

PLANNING, STEPS AUR EK SHARED IDEA

1

PEHLE QAFIYA, PHIR SHER

PLAN AHEAD

Do line ki poem mein, jahan line "grab it" par khatam honi thi, model ne line likhne se **pehle** rhyme wala word "rabbit" chun liya.

Researchers ne "rabbit" ko dabaya to line "habit" par khatam hui. "green" daala to line "green" par khatam hui.

Matlab: model aage ka target chunta hai, phir line us tak banata hai. Aur use badla bhi ja sakta hai.

2

DALLAS SE AUSTIN TAK

MULTI-STEP REASONING

Sawaal: "Dallas jis state mein hai, uski capital?" Andar pehle "Dallas Texas mein hai" activate hota hai, phir "Texas ki capital Austin hai".

Researchers ne Texas ki jagah California daala to jawab Austin se badal kar **Sacramento** ho gaya.

Matlab: beech ka step sach mein jawab chalata hai. Model facts jodta hai, sirf ratta hua jawab nahi dohrata.

3

LANGUAGE ALAG, IDEA EK

SHARED CONCEPT SPACE

"Small ka opposite" alag-alag languages mein poochha gaya. Chhota hona aur opposite hona wale wahi core features activate hue, phir "bada" ka idea question ki language mein translate hua.

Authors ke mutabiq Claude 3.5 Haiku ke features ka kaafi bada hissa languages ke beech shared hai, ek chhote model se do guna se zyada.

Matlab: andar ek common "soch" ki jagah hai, language baad mein lagti hai.

MATH, BHAROSA AUR HALLUCINATION

4

DO RASTE EK SAATH

MENTAL MATH: 36 + 59

Model ne do raste ek saath chalaye: ek mota **andaaza** nikalta hai, doosra **last digit** sahi nikalta hai. Dono mil kar 95 bante hain.

Par jab poochha "95 kaise aaya", model ne school wala **carry the 1** method bataya, jo andar ki asli strategy se match nahi karta.

Matlab: model ka diya hua explanation, uske asli kaam ka bayaan nahi bhi ho sakta.

5

JO BOLA, WAHI KIYA?

FAITHFUL VS UNFAITHFUL REASONING

0.64 ka square root: model ke likhe steps andar ke asli step (64 ka square root) se match hue.

Ek bade number ke cosine mein model ne "calculation" karne ka daava kiya, par researchers ko aisa kuch hota hua **nahi mila**. Hint dene par kabhi-kabhi wo hint wale jawab tak ulta kaam karke pahunch gaya, jise authors "motivated reasoning" kehte hain.

Matlab: likha hua reasoning hamesha sachcha record nahi hota.

6

JAANA-PEHCHANA NAAM, GALAT JAWAB

HALLUCINATION

Default mein model "mujhe nahi pata" bolta hai. Jab "known entity" feature chalta hai (jaise Michael Jordan), wo default dab jata hai aur model jawab deta hai. Anjaan naam (Michael Batkin) par model mana karta hai.

Galti tab hoti hai jab naam **jaana-pehchana lage** par us bare mein kam pata ho: tab model andaaza se kahani bana deta hai.

Matlab: half-known cheezon par AI ke jawab ko khaas check karo.

LIMITS AUR AAPKE LIYE

AUTHORS KHUD KYA MAANTE HAIN

- Ye tareeka model ke total computation ka sirf **ek hissa** pakadta hai, chhote aur simple prompts par bhi.
- Kuch mechanisms tools ki wajah se bane artifacts ho sakte hain, model ki asli soch nahi.
- Sirf kuch dozen words ke prompt ko samajhne mein bhi insaan ke kuch ghante lagte hain.

REEL MEIN KYA SIMPLE KIYA

REEL MEIN	SAHI TASVEER
"Claude" aur "AI"	Deep studies Claude 3.5 Haiku par hain. Dusre models par aisa hi hoga, ye is research se sabit nahi hota.
"Andar jhaanka"	Ek metaphor hai. Researchers ne features aur circuits trace kiye, model ka "dimaag padha" nahi.
"Plan banata hai"	Rhyming poem ke case mein dikha. Har tarah ke kaam mein planning ka daava nahi hai.

MERA SUGGESTION: AI KO AISE USE KARO

- AI ke **explanation** ko uske kaam ka saboot mat samjho. Answer alag cheez hai, explanation alag.
- Half-jaane naam, niche topics aur numbers par **source maango** aur khud check karo.
- Zaroori kaam mein final jawab ek aur jagah se verify karo.

CHHOTA GLOSSARY

Feature	Model ke andar ka ek idea ya pattern, jaise "Texas" ya "opposite".
Circuit	Features ki chain jo milkar ek jawab banati hai.
Chain of thought	Model ke likhe hue step-by-step reasoning.
Hallucination	Model ka confidence ke saath galat ya banaya hua jawab dena.

Source: Anthropic, "Tracing the thoughts of a large language model"

Aur aise breakdown chahiye? Follow karo **@techie.addy**.